

Project Initialization and Planning Phase

Date	4 JULY 2026
Team ID	SWUID2026215775
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop on ai and ml based on the credit card approval prediction system that analyzes customer information such as income employment status, credit card application should be approved or rejected
Scope	The project focuses on building intelligent prediction system for credit card approval it includes customer data collection data preprocessing machine learning and model training and result visualization.
Problem Statement	
Description	Traditional credit card approval methods on manual verification, which is time consuming error-prone and inconsistent. Banks often need to analyze multiple customer attributes such as income, employment credit history and existing liabilities before making a decision this process delays Approval and affects customer experience
Impact	An automated ai based prediction system improves the speed and accuracy of credit card approval decisions it reduces manual effort minimizes human errors enhances customer satisfaction and helps financial institutions make reliable and data driven decisions
Proposed Solution	
Approach	Develop a machine learning model that analyzes customer details financial information employment status and credit history to predict credit card approval the system process the input data validates it performs prediction using trained ML algorithm and displays an instant approval or rejection result

Key Features	AI&ML based prediction fast and accurate approval decisions secure customer data processing
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	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor required to train and run the ml model	Intel i5/ryzen 5 processor
Memory	RAM specifications for data processing and model execution	8 GB RAM
Storage	Disk space for data, models,logs,data sets and project files	256 GB SSD or above
Software		
Frameworks	Python frameworks used to build the web applications	Flask
Libraries	Libraries used for data processing,model training and prediction	scikit-learn, pandas, numpy, matplotlib, seaborn,joblib
Development Environment	Tolls used to write test and run the project	Jupyter Notebook,VSCODE
Data		
Data	Source of customer application and credit-related data used for training and testing the ML mode	Credit card approval dataset(csv format)